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Use of multi-agent system for industrial production control

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Abstract: This paper describes specific implementation of Asset Administration Shell technology used for an industrial distributed control system. The base of this control system is a multiagent system of digital twins which was created by AAS. This multiagent system was developed for a specific platform CP Factory (mainly based on Simatic PLC) but the implementation is y portable to each common platform. Digital twin of several types of components of typical industrial production process are described in this paper with the implementation of active and passive administration shell. The main focus is on describing the way of integrating digital twins into the production system and the necessary SW tools. Industry 4.0 language was used as a common communication protocol as well as OPC UA and MQTT.

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Keywords: Multiagent system, Asset administration shell, digital twin, OPC UA, MQTT

1. INTRODUCTION

The terms Industry 4.0, digitalization or virtualization have been used frequently for some time. In recent years, these principles have finally begun to be applied in real industrial applications. Here, the biggest challenges are the need for standardization, unification of communication interfaces and openness of communication protocols, while maintaining the maximum requirements for security and reliability of the solution. Many solutions have already been theoretically described and tested to a limited extent, but the challenge remains to deploy these technologies on truly large-scale industrial systems consisting of tens or hundreds of components from different producers and to confirm the effectiveness of these solutions in a truly distributed industrial production control system. This paper describes the design of a solution that has the ambition to become an exemplary application of these technologies in a system that is very close to a real industrial environment.

It is obvious that the Asset Administration Shell technology has the potential to become, at least at the European level, a unified standard for the description of digital and communication interfaces of Industry 4.0 components (the term component in this case is actually understood according to the specification I4 components according (VDI/VDE, 2015)), the creation of so-called digital twins and their mutual interaction. The article aims to present the use of this standard for the creation of a multi-agent control system, implementing a distributed industrial production control system. This system contains a complete production process starting from the creation of a customer order through the specification of the production process and, above all, the dynamic allocation of production resources. In contrast, the text does not deal with specific algorithms for the selection of the ideal provider of each production operation. This topic is in itself very complex and provides scope for independent research.

2. THE HW PLATFORM DESCRIPTION

The solution described in this article was developed and deployed on FESTO's CP Factory platform. It is a cyber-physical manufacturing system developed primarily for teaching and demonstrating the principles of Industry 4.0. The advantage of this system is that it is composed of real components used in industry such as Simatic PLCs, electric and pneumatic actuators and valves, optical components or RFID. The system also supports easy virtualization of individual components and the interconnection of virtual and physical units and their interaction with each other.

The basis of the platform is a physical production line consisting of conveyor belts, handling and assembly robots, production stations (drill, press, inspection station, etc.) and transport robots such as AVG Robotino.

The original CP Factory line control system uses the classic TPC protocol for communication between the modules and the

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MES system, which transmits simple binary data. This system was found to be inadequate for the requirements of Industry 4.0 according to (Marcoň et al., 2022) This was mainly due to the lack of dynamic allocation of production resources. The MES4 software, which was part of the original solution, functioned mainly as a bridge between the communication protocol of the production modules and SQL queries to the database. For the purpose of this paper, a completely new solution was designed, which retains only the numerical labelling of individual production operations and semi-finished products from the original system. The reason for using the same marking is only to make both systems easier to use.

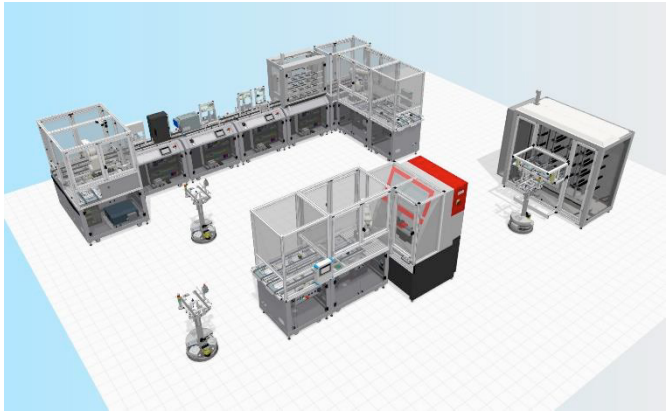


Figure 1. 3D model of actual HW configuration of CP Factory – HW testing platform for multiagent system

3. MULTIAGENT SYSTEM

The distributed manufacturing process control system described in this paper is based on a multi-agent digital twin system. Each agent is implemented by a digital twin of a specific component of the manufacturing process. Each digital twin is then designed according to the specification of the AAS standard (Platform Industrie 4.0, 2022). All twins are designed to be proactive in terms of this specification and are therefore capable of independently initiating communication with other agents - digital twins of other I4.0 components.

Individual agents consist of a passive and an active part of the digital twin of a given I4.0 component. The passive part, depending on the component type, consists of a static XML or JSON file or data located on the OPC UA server. The active part of the digital twin is always made up of a separately running python program. These programs are actually run as separate processes from the operating system's point of view and do not use any shared data or memory. Communication between the agents is provided by the Industry 4.0 language.

Individual agents are able to independently log in and out of the system and communicate with surrounding agents, ensuring the implementation of individual production operations. For this negotiation, the multi-agent system does not need any runtime platform in the sense of an application providing their mutual communication or allocation of production or communication resources or time slots. Everything is provided only by the communication of the individual agents with each other.

The system needs only one shared entity to run, which is the System Admin SW module that ensures correct agent login and logout to this system. This login means both authentication of a given agent and then, above all, assignment of a unique identifier of this agent in the whole system and information about the identifiers of other agents. The address range on which the other running agents occur is passed. If necessary, information is also sent to all other agents about changes to this address space range - i.e. the range of addresses that can be assigned to specific agents. Agent logout from the system is done similarly. The agent's identifier is released and, if necessary, the range of address space reserved for agents of a particular type is reduced.

System Admin is the only central element of the whole system. However, it is not tightly coupled to the individual agents and does not act as a central element in the sense of controlling the production process or the runtime environment for the individual digital twins. It functions only as an active database of these digital twins, in which the unique identifiers of these agents are stored together with their network addresses. System Admin is able to send these agents a message containing information about system-wide changes (typically a change in the address range for a particular group of agents), but is not able to create, terminate or issue commands to these agents independently. From an operating system perspective, individual agents and System Admin are completely independent, self-running programs.

According to the conclusions published in (Kaczmarczyk et al., 2022), it follows that even in a distributed control system the existence of a central element is appropriate or even necessary. For use-cases such as the evaluation of KPIs or other production efficiency parameters, the existence of a central element capable of collecting data from all components of the entire system is always preferable. In line with this idea, System Admin can be part of just such a central element, which will be built at the level of the MES or ERP system. However, this is not strictly necessary for its functionality.

An often mentioned advantage of a distributed control system is its flexibility and ability to respond dynamically to changes in the environment, such as an outage or the addition of a new production component. This is of course also true in this case. However, the question arises whether the same result could not be achieved by the effective use of artificial intelligence algorithms in a centrally controlled system.

However, it is clear that a distributed system of autonomously operating agents is more fault-tolerant than a centrally controlled system. There is no possibility of a failure of the central node that would bring the entire production to a halt and possibly result in data loss. In this respect, the distributed system is clearly more robust and stable.

4. SW MODULE TYPES

If we understand manufacturing operations as services, we are able to define two basic types of components. Service requester - a component that requests a certain operation of the production process and service provider - a component that provides this operation. We are able to apply this basic division to all parts of the production process, starting with the ordering of a product and ending with its shipment to the customer.

In our particular case, we distinguish several specific types of production process entities, which correspond to several types

of agents, i.e., digital twins of these components. However, it is always possible to view a given agent type as either a service provider or a service requester. In addition, there are SW modules that are not digital twins of I4.0 components and serve only as subsystems providing certain functionality.

- Product or semi-finished product
- Ordering system (E-shop)
- Product Manager
- Production resource
- Transport element
- System Admin

4.1 Ordering system

This part of the system consists of a classic e-shop, which is separated as much as possible from the rest of the digital twin system. The reason for this separation is mainly the possibility to implement web frontend and backend technologies in standard ways without the need to know the AAS specification and the digital twin communication system. Here, user order parameters such as price, delivery date, or optional product parameters are organized into a JSON data package and passed to the Product Manager with a *CallForProposal* message. The MQTT protocol is used for this transfer but can be easily replaced by OPC UA or another communication standard.
This part of the system is therefore not a digital twin in terms of the AAS specification, nor does it directly use this technology.

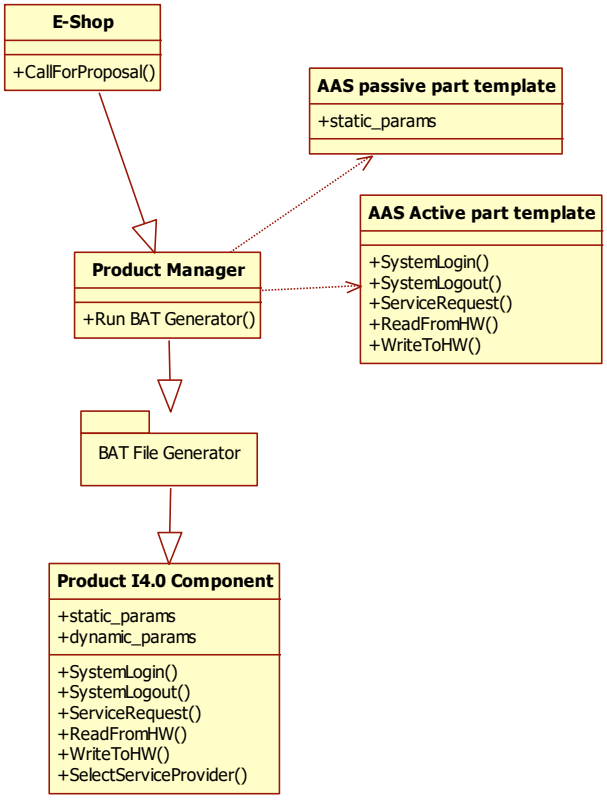


Figure 2 – Generating final product I4.0 component from active and passive templates

4.2 Product Manager

In this part of the system, digital envelopes of ordered products are created. Templates for passive and active parts of product type AAS are prepared here, containing the division and description of each submodule. According to these templates and the data passed from the e-shop, specific digital twins of target products are generated. The actual execution of the prepared agent is done by means of a batch file, which ensures execution in the form of a self-running process. From this point on, there is no connection between the running agent and the Product Manager. The Product Manager is unable to interact with the agent or terminate it in any way. In this implementation, the Product Manager is not a digital twin of any particular I4.0 component. It is a software module that creates and starts products AAS. In principle, it would be possible to implement this SW as a digital twin of the e-shop itself, but in this implementation, it was chosen to separate these two entities from each other as much as possible.
The templates of the passive and active parts are designed according to the specification (Platform Industrie 4.0, 2022) and their implementation is based mainly on the description of the demonstrator I4.0 Language (Belyaev, 2019)

4.3 System Admin

This part of the system is not a digital twin of any Industry 4.0 component, nor is it a separate agent. It is a central database of all running agents, which stores information about their identifiers and network addresses. From an operating system perspective, it is a standalone running program. The presence of this central element is necessary to allow the smooth integration of a new component into the system. The newly created agent has no information about which other agents exist in the system and how it can communicate with them. Given a truly physically distributed production, it is not possible to rely on all components of the production process to run in a single closed internal network.
Therefore, the condition for a new agent to join the system is that this agent is equipped with information that allows it to send a login request to this element. In response, System Admin assigns it a unique identifier and information about the range of network addresses where other agents can be found. This ends the connection between the agent and System Admin. The disadvantage of this state is that the central element is not able to recognize a state where the agent is terminated due to an error, network or power failure. From the perspective of the central element, the agent will still exist in the system, an identifier will be reserved for it and its address will be known. This fact would introduce the possibility of error, the existence of zombie agents, into the solution. The solution is to introduce periodic heartbeat queries with a very long period (tens of minutes). This way the system is not

overloaded with unnecessary periodic queries and in the long run dead agents are released from the database.
It should be emphasized that System Admin is not a runtime platform or environment for running a multi-agent system. It works only as a database of initialization information necessary to integrate a newly created agent into the system. Its failure or malfunction will have no effect on the running of the production process. All already ordered products that have generated their AAS will continue to be produced without any change. However, a long-term failure of this subsystem will make it impossible to place new orders.

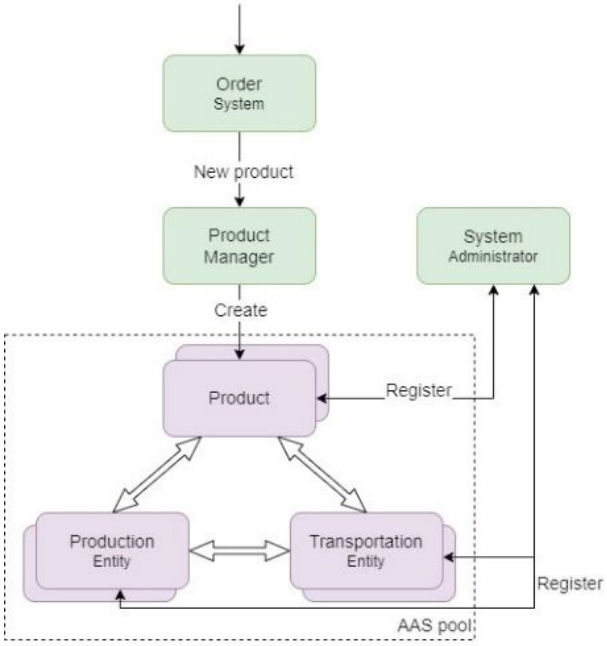


Figure 3. Diagram of interactions of software components of the system

4.4 I4.0 component - product

The product or semi-finished product entity is an I4.0 component due to the existence of a digital twin containing complete information about the production process of the product, the current production status and the customer parameters of the product (order identification, price, delivery date, etc.).
In this case, the passive part of the administrative shell consists of a JSON file containing the aforementioned information.
It is necessary that a component of type product is a proactive element. Only in this way is it able to reach out to components of the production resource type and negotiate the execution of the production operation. It is therefore clear that this type of agent is of the Service requester type. The active part of the administrative shell is implemented by Pyton script. This script contains functions for:

- Reading and writing to the passive part
- Functions for communication with the physical component (see below)
- Login and logout to the entire production system (login to System Admin)
- Communication with agents of the production resource type to request and confirm the execution of a specific production operation
- Communication with agents of the transport element type for request and confirmation of a transport operation
- Algorithms for selecting the best supplier for the requested operation

In this case, the functions for communication with the physical component are strongly implementation dependent on the HW platform used. When the system is deployed on the CP Factory platform, the individual semi-finished products are not equipped with any communication interface for unambiguous identification or communication with their digital twin. Here, this connection is ensured by means of transport carriers on which the individual semi-finished products are placed during the production process. These are already equipped with an RFID chip that is able to provide the necessary data link of the component to its administrative shell. Thus, the functions for communicating with the physical component in this case are actually functions for reading and writing from/to the passive part of the transport element type agents. If the system is deployed on a different platform, these functions will need to be modified accordingly.

4.5 I4.0 component – operation resource

Production resources, i.e. workstations or robots, are I4.0 components as there is a clear digital link to their administrative shell. In the CP Factory system, all such elements are equipped with the OPC UA communication interface and therefore it is the data provided by the physical unit via this interface that is considered as a passive part of the administrative shell. This approach was chosen mainly to simplify the implementation, where the need to copy this data to an external file and keep this file up-to-date disappeared.
The active part of the administrative shell is again made up of Python scripts while the specific program module is generated from one universal template depending on the data from the static part. For this type of component, unlike products, one cannot assume too dynamic increase or decrease of attached units. Thus, there is no need for a manager that generates specific digital twins automatically. In the case of a new production resource being connected to the system, the generation and execution of the agent in question must be done manually. The active part contains functions for:

- Login and logout from/to the entire production system
- Communication with the passive part via OPC UA
- Receipt of production operation requests
- Communication about the conclusion of a contract with a product agent
- Cancellation/suspension of a contract with a product agent

At first glance, it might seem that this type of agent could be just a reactive digital twin that will only respond to the demand

of a given service. However, in the proposed system we also consider a behavior where the agreed service between the requester and the provider will be deferred by the provider. The reason for this behavior is the demand of another requester that requests the service with higher priority. In other words, he is able to "pay" for the priority handling of his request. In such a case, the service provider needs to proactively establish communication with the original service requester and inform it of the change in the expected service delivery date. For this reason, this type of administrative shell must also be designed to be proactive from the beginning.

4.6 I4.0 component – transport module

The transport operation can, with a certain degree of generalization, be understood in the same way as a classical manufacturing operation. However, due to the different character of these operations (transport does not change the state of the semi-finished product) and also with regard to flexibility in deployment on different HW platforms, it is advantageous to separate the transport elements as separate components. However, they are clearly components in the sense of I4.0. There is a clear data link to the corresponding administrative shell, which is formed (in the case of the CP Factory HW platform) by the OPC UA communication interface. This provides direct access to the data of the physical element, whether it is just a transport carrier, equipped with an RFID chip, or an intelligent AGV Robotino.

Thus, the passive part of the administrative shell is also here made up directly of the data stored in the physical module. The active part is then made up of Python functions, similarly to the case of a component of the production resource type. The meaning of the individual functions of this part is the same as in the previous case, but their implementation is adapted to the specifics of the transport operations.

Also here it is suitable to consider from the beginning the use-case when the transport module postpones the already arranged service due to the acceptance of an order with a higher priority. Here, too, it is necessary to implement the administrative shell as proactive. From the above description it is also clear that this type of agent acts as a service provider in the system.

6. CONCLUSIONS

This paper describes the design of a multi-agent system according to the AAS specification, intended for an autonomous distributed control system. The focus is mainly on the description of the different types of components involved in the manufacturing process and the corresponding types of digital twins. The main part of the work focuses on the system for generating active and passive twin parts from templates and managing their life cycle. The software modules necessary for automatic generation of these twins and for their registration into the running production system are presented. This process, as shown, cannot be implemented without the existence of a central element that serves as a database of identifiers and addresses of the currently running agents. The importance of this central element is limited to a maximum in the described solution. It does not participate in any way in the mutual communication between the agents nor does it interfere in the execution of the individual steps of the production process. The described application represents a complex and efficient solution for the creation of digital twins and their automatic integration into the production system. The process of selecting the ideal supplier of the desired service is a separate task that will be addressed in the follow-up research. This process, which at the current stage of development is limited to a simple comparison of the price of the requested service, can realistically be a very complex problem. The selection of a supplier can be made not only according to a considerable number of parameters, but also with a view to several subsequent steps or the overall efficiency of the whole system. The authors of the described solution have made every effort to design the whole system from the very beginning to be ready for these complex tasks.

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